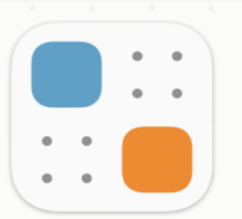


Patch Pure DATA



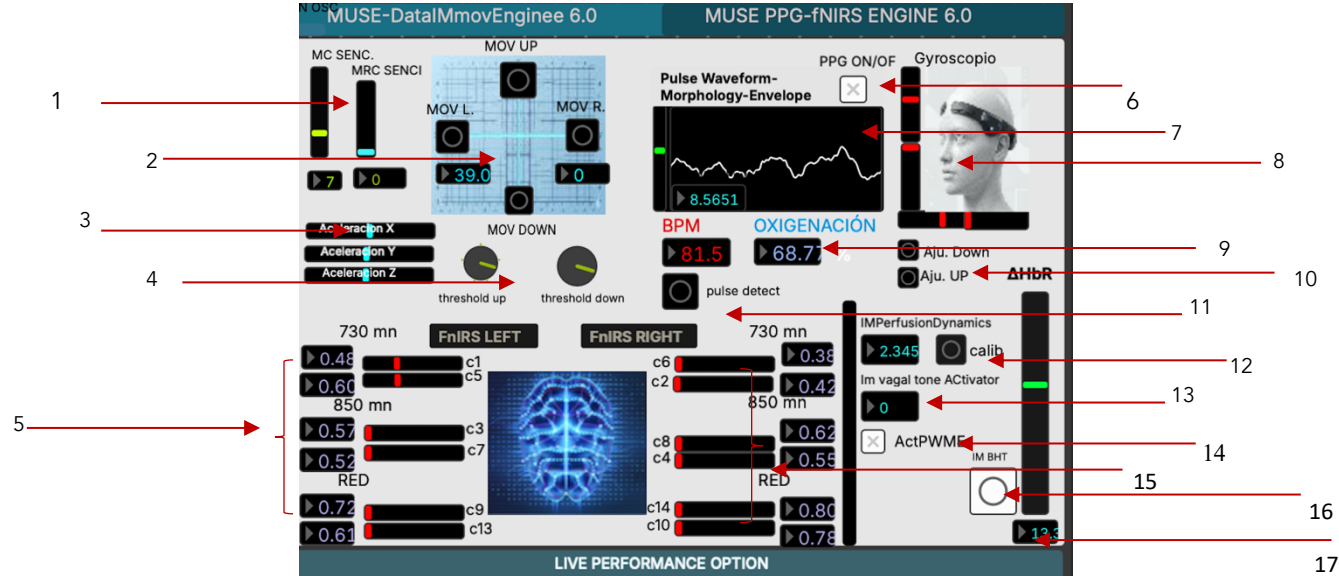
PLUG DATA

MUSE-DATAIMMOV FNIRS V6.0 MODULE

Diseño y desarrollo de sistemas MMHCI híbridos con bioseñales y un DMI de smartphones, para obras bio-interactivas mixtas y performáticas

Pure Data Patches for the PhD Thesis: Juan Pablo posada Alvarez





Motion recognition with MuseAthne is intrinsically integrated into head movement. This is based on the position and acceleration measurements captured by the accelerometer and gyroscope integrated in the device. The algorithms are designed to identify four discrete and four continuous states that correspond to possible tilts of the head in different directions. To achieve this, the algorithm must estimate when the head is tilted in one of the four predetermined directions, using only the data from the two-axis accelerometer: x. y

Detection thresholds are configured to act as binary gates, aligned with the acceleration values corresponding to each movement.

Here's a general guide to module functions and connections in PD:

1. **Accel and Giro filter:** These sliders regulate the parameters of the moving average filter applied to the signals of the accelerometer and gyroscope of the MUSE module, used. Adjusting the filter window allows you to:
 - Low values: They maintain the original signal including noise components.

- High values: Reduce noise but increase system latency. The balance between signal stability and response time must be optimized according to the requirements of each construction site
2. **IM MRC plane (x,y):** A 2D Cartesian visualization that maps MRC interactions, where each quadrant corresponds to a direction of movement.
 3. **Accel and Giro Data:** This display graphically represents the values of the accelerometer and gyroscope sensors in the three axes (x, y, z), with the following characteristics:
 - Measuring range: $\pm 9.81 \text{ m/s}^2$ (standard gravitational acceleration).
 - Center point: Indicates zero acceleration.
 - Extremes: They show positive (top) and negative (bottom) acceleration.
 4. **V.up and V down threshold:** A parameter that controls the detection threshold for upward movements in MRCs, with the following specifications:
 - Default: 120.
 - Adjustable range: 70-120.
 - Values close to 70 increase the sensitivity of the system.

The V down slider is opposed to the previous threshold for downward movements with the same range (70-120) and default value (120). Reducing the value improves detection sensitivity.

5. fNIRS (functional Near-Infrared Spectroscopy) is an optical sensing technique that uses near-infrared light to detect changes in blood oxygenation and hemodynamic activity in the tissue beneath the sensors. In the Muse Athena, the 730 nm optical channels are organized in Inner/Outer pairs corresponding to sensors C1–C5, C3–C7, C9–C13 and their respective paired channels.

This section adjust the Baseline Offset: A parameter that adjusts the baseline offset of the 730 nm Inner/Outer optical signals according to variations in the position and contact pressure of the Muse Athena on the forehead.

The parameter allows compensation for baseline variations caused by changes in sensor position and contact pressure, maintaining the optical signals within a suitable operating range for subsequent electronic mapping.

6. **PPG Activation:** Enables or disables the real-time processing and mapping of the optical signal used to extract the cardiac pulse wave.

OFF: The PPG processing system remains inactive.

ON: Activates real-time PPG signal processing and makes the pulse-wave signal display active

7. **PPG Waveform Display:** Displays the real-time pulse-wave signal extracted from the Muse Athena optical sensors.

The display provides a visual representation of the temporal variations of the PPG signal, allowing the user to monitor the cardiac pulse waveform and verify its stability during real-time interaction.

8. **MCC Visualizer:** Two-dimensional representation (x, y axes) of the cephalic position, with the following characteristics:

- Central point: Neutral position.
- Extreme: Indicate positive/negative displacements.
- Graduated markers: Facilitate precise adjustment of ranges of motion

9. **Oxygenation and BPM Display:** Displays the estimated blood oxygenation level and heart rate derived from the optical signals of the Muse Athena.

The display provides real-time numerical values of oxygenation (SpO₂) and heart rate (BPM) for monitoring and interaction purposes within the electronic system. These values are intended exclusively for artistic, experimental, and interactive applications and are not intended for medical diagnosis, monitoring, or clinical use.

10. **Up/Down Adjustment:** Button that initiates the calibration protocol for the MCC Y-axis. During the process, an auditory guide indicates the steps required for optimal system adjustment.

11. **Pulse Peak LED:** Provides a visual indication each time a cardiac pulse peak is detected in the PPG signal.

The LED is triggered in real time by the detection of the pulse-wave peak, providing an immediate visual representation of the cardiac rhythm for interactive and artistic purposes. It is not intended for medical diagnosis or clinical monitoring.

12. **Baseline Calibration:** Establishes an individual baseline reference for the fNIRS optical signals measured by the Muse Athena.

When activated, the system records the current optical signal levels and stores them as reference values for subsequent processing and interaction. This allows the system to compensate for individual differences and variations in sensor placement and contact pressure during the performance.

The calibration is intended exclusively for artistic, experimental, and interactive purposes and is not intended for medical diagnosis or clinical monitoring.

13. **Perfusion Dynamic:** Displays the real-time dynamic variation of the fNIRS signal associated with changes in blood perfusion during controlled breathing and breath-hold events.

IM BHT (Breath Hold Trigger): Detects significant changes in the perfusion-related signal and generates an interaction event when the predefined variation associated with a breath-hold condition is reached.

The resulting values are used as a control signal for real-time electronic mapping, allowing changes in the physiological state to influence the behavior of the electronic system. These values are intended exclusively for artistic, experimental, and interactive purposes and are not intended for medical diagnosis, monitoring, or clinical use.

14. **Pulse Envelope — MIDI Output**

MIDI Output: Enables or disables the transmission of the pulse-wave envelope extracted from the Muse Athena PPG signal via MIDI.

OFF: MIDI transmission of the pulse envelope is disabled.

ON: The real-time pulse envelope is transmitted via MIDI and can be used as a control signal for external audio and electronic processes.

15. This function allows the cardiac pulse dynamics to be mapped in real time to musical and electronic parameters. The MIDI output is intended exclusively for artistic, experimental, and interactive applications and is not intended for medical diagnosis or clinical monitoring.
16. In the Muse Athena, the 850 nm optical channels are organized in Inner/Outer pairs corresponding to sensors C2–C6, C4–C4, C12–C14 and their respective paired channels.

This section adjust the Baseline Offset: A parameter that adjusts the baseline offset of the 850 nm Inner/Outer optical signals according to variations in the position and contact pressure of the Muse Athena on the forehead.

The parameter allows compensation for baseline variations caused by changes in sensor position and contact pressure, maintaining the optical signals within a suitable operating range for subsequent electronic mapping.

17. **IM BHT — Breath Hold Threshold LED**

Threshold LED: Provides a visual indication when the perfusion-related signal reaches or exceeds the predefined threshold of the IM BHT (Breath Hold Trigger) algorithm.

The LED is activated in real time when the threshold condition is detected, providing a visual confirmation of the breath-hold event for the interactive electronic system.

The indicator is intended exclusively for artistic, experimental, and interactive applications and is not intended for medical diagnosis, monitoring, or clinical use.

18. **Deoxygenated Hemoglobin (HbR)** Displays the real-time variation of the estimated deoxygenated hemoglobin signal obtained from the Muse Athena fNIRS sensors.

The slider provides a visual representation of the dynamic changes in HbR associated with variations in blood oxygenation during breathing. The displayed value typically increases during deep inhalation and breath-hold periods and decreases as the air is released, providing a real-time control signal for interactive electronic mapping.

The displayed values are intended exclusively for artistic, experimental, and interactive applications and are not intended for medical diagnosis, monitoring, or clinical use.

The module is connected as follows:

Outlet 1-4: output of y-axis (positive/negative) and horizontal x-axis (positive/negative) MCC IM values in MIDI values. Each output provides values in a range of 0 to 127 for each of the 4 sections of the head movement axes.

Outlet 5-8: Output of MRC IM values from the axis (x, y) in MIDI values. Each output provides values in a range of 0 to 127 for each of the 4 sections of the head movement axes.